Homework 2
Network Dynamics and Learning

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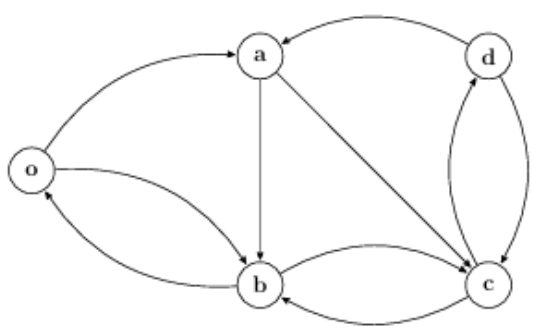
Exercise 1: Continuous-Time Random Walks and DeGroot Opinion Dynamics

$$\Lambda = \begin{array}{c} \begin{array}{ccccc} o & a & b & c & d \end{array} \\ \begin{pmatrix} 0 & 2/5 & 1/5 & 0 & 0 \\ 0 & 0 & 3/4 & 1/4 & 0 \\ 1/2 & 0 & 0 & 1/3 & 0 \\ 0 & 0 & 1/3 & 0 & 2/3 \\ 0 & 1/3 & 0 & 1/3 & 0 \end{pmatrix} \end{array} \begin{array}{c} \\ o \\ a \\ b \\ c \\ d \end{array} .$$



Part a

What is, according to the simulations, the average time it takes a particle that starts in node a to leave the node and then return to it?

To estimate the average return time to a given node in the network, we simulate a continuous-time random walk using the transition rates derived from the matrix Λ. After computing the total leaving rates and determining the maximum rate $w_\star$, we construct a stochastic transition matrix Q through the uniformization method. This matrix Q specifies the probabilities of moving from one node to another at each jump.

In each simulation, the particle starts at the chosen node. The time until the next transition is sampled from an exponential distribution with rate $w_\star$,

$$t_{\text{next}} = -\frac{\ln(u)}{w_\star}, \qquad u \sim \mathcal{U}(0,1),$$

and the next node is selected according to the corresponding row of Q. The particle continues to move through the network in this way until it returns to the starting node. The total time elapsed during this path is recorded.

This procedure is repeated many times (10,000 simulations). The output provides the average time for a particle to return from a specified start node to the same node, giving insight into the network's dynamics and the efficiency of information dissemination or particle movement within the network

Average return time to node 'a': 6.711

Part b

How does the result in part (a) compare to the theoretical return time $\mathbb{E}_a[T_a^+]$? (Include a description of how this is computed.)

First, the stationary distribution π of the chain is computed by finding the eigenvector of $Q^\top$ associated with the eigenvalue 1 and normalizing it so that its entries sum to one. The theoretical mean return time to node a in a continuous-time Markov chain is then given by

$$\mathbb{E}_a[T_a^+] = \frac{1}{\pi_a\, w_a},$$

where π_a is the stationary probability of node a and w_a is the total rate of leaving node a. Finally, this theoretical value is compared with the simulated return time by computing their absolute difference. After computing pi and pibar I calculated it. the theoretical return time is equal to 6.708, which is almost the same as part a.

Theoretical return time to node 'a': 6.708
Difference between simulation and theory: 0.0003

Part c

What is, according to the simulations, the average time it takes to move from node o to node d?

In Section C, the focus shifts to computing the average time it takes for a particle to travel from one specific node (node 'o') to another (node 'd') in the network. This is achieved by adapting the same simulation technique used in Section A but with different start and end nodes. The objective is to understand the connectivity and efficiency of paths between different nodes within the network. The output shows the average time taken for a particle to move from node 'o' to node 'd', highlighting the network's structure and the influence of different paths on the movement of information or particles.

Average time from node 'o' to 'd': 10.7372

part d: How does the result in c compare to the theoretical hitting-time $\mathbb{E}_o[T_d^+]$? (Describe also how this is computed.)

Part d

Section D involves calculating the theoretical hitting time from node o to node d, using the same method applied in Section B. The linear system defined by the modified transition matrix $\hat{P}$ and the leaving rates $\hat{w}$ is solved to obtain the expected hitting times. The theoretical value is given by

$$\mathbb{E}_o[T_d] \;=\; (I - \hat{P})^{-1}\,\mathbf{1}/\,\hat{w},$$

where $(I - \hat{P})^{-1}$ is the fundamental matrix and $\mathbf{1}$ is a vector of ones. The result is then compared with the simulated hitting time from Section C, allowing us to evaluate the accuracy and reliability of the simulation model. The output includes the computed theoretical hitting time and the simulation error, offering insight into the effectiveness of the paths between the nodes in the network.
Calculated hitting time (o to d): 10.7666
Simulation error: 0.02942

Part e

Does the graph G defined by Λ (where an edge (i, j) exists if $\Lambda_{ij} > 0$) satisfy the conditions for consensus to be achieved in the French-DeGroot model? If so, simulate this dynamics on G: do the opinions converge? (For clarity of visualization, run more than one experiment on different initial conditions, and plot the evolution of the nodes' opinions in each experiment.)

In this exercise, we analyzed whether the graph G, defined by the weight matrix Λ, satisfies the conditions necessary for achieving consensus in the French-DeGroot model. The conditions checked were:

- The graph must have only one strongly connected component (sink in the condensation graph).
- The sink component must be aperiodic.

After analyzing the structural properties of the graph G, we found that:

- The graph G has only one strongly connected component, meaning it satisfies the first condition.
- The sink component is aperiodic, ensuring the second condition is also satisfied.

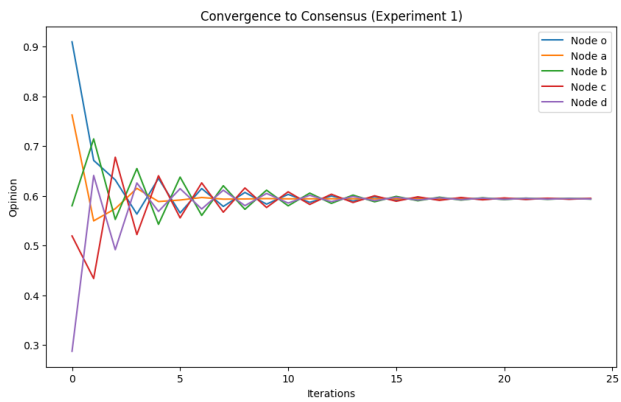
To further study the opinion dynamics, we simulated the French-DeGroot model on G. First, we computed the adjacency matrix A and the degree matrix D of G. For nodes with no outgoing edges, we adjusted the degree matrix to avoid singularity. The transition probability matrix P was calculated as:

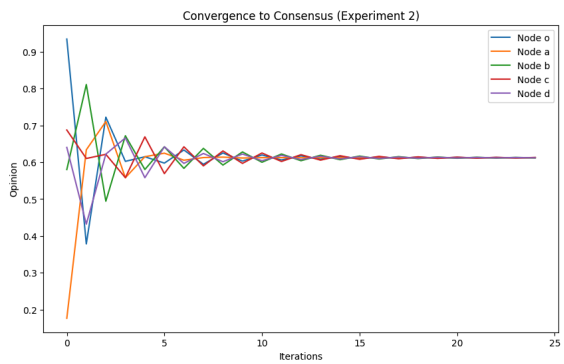
$$P = D^{-1} \cdot A.$$

In the simulation, the opinions of nodes were initialized randomly, and the dynamics evolved according to:

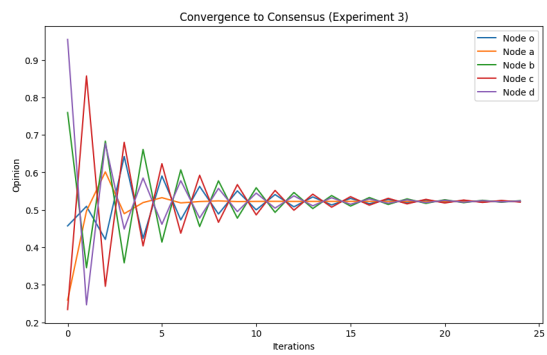
$$x(t+1) = P \cdot x(t),$$

where $x(t)$ represents the opinions of nodes at iteration t. The simulation was repeated for multiple experiments, and the final opinions of the nodes were analyzed to check for convergence.





(a) Experiment 2



(b) Experiment 3

Figure 1: Simulation results for G across two experiments. Both experiments show that the opinion dynamics converge to a consensus state.

The results, illustrated in Figure 1, showed that the opinions converged to a single value. This is because:

- The graph G satisfies the conditions for consensus: having one strongly connected component and an aperiodic sink.
- The convergence reflects the consensus value achieved by all nodes in the network.

Thus, the French-DeGroot dynamics on G successfully led to consensus under the given conditions.

Part f

Simulate the dynamics many times (≥ 200) to estimate the variance of the consensus value. The initial opinions are assumed to be i.i.d. according to $x_i(0) \sim \mathcal{N}(0, \sigma_i^2)$ where $\sigma^2 = [1, 2, 2, 2, 1]$. Compare this value with the theoretical result

$$\mathrm{Var}(\alpha) = \sum_{i=1}^{n} \pi_i^2 \sigma_i^2.$$

In this part of the exercise, we aim to compute and compare the theoretical variance of the consensus value with the simulated variance obtained from simulations. The process involves the following steps:

Theoretical Variance

The theoretical variance of the consensus value is derived analytically using the properties of the graph and the stationary distribution. Let the stationary distribution of the graph be represented as $\pi = [\pi_1, \pi_2, \ldots, \pi_n]$, which is the eigenvector of the transpose of the transition probability matrix P, corresponding to the eigenvalue 1. The stationary distribution satisfies:

$$P^T \cdot \pi = \pi, \quad \text{and} \quad \sum_{i=1}^{n} \pi_i = 1.$$

Given the variances of the initial opinions σ_i^2 for each node i, the theoretical variance of the consensus value α is computed as:

$$\mathrm{Var}(\alpha) = \sum_{i=1}^{n} \pi_i^2 \cdot \sigma_i^2.$$

The logic behind this formula is that the consensus value α is a weighted average of the initial opinions, where the weights are determined by the stationary distribution π. Since the initial opinions are random variables with variances σ_i^2, the variance of the weighted average is given by $\sum_{i=1}^{n} \pi_i^2 \cdot \sigma_i^2$.

In the implementation, the variances for the nodes are represented as:

$$\sigma^2 = [\sigma_o^2, \sigma_a^2, \sigma_b^2, \sigma_c^2, \sigma_d^2] = [1, 2, 2, 2, 1].$$

The stationary distribution π is computed from P^T by normalizing the eigenvector corresponding to the eigenvalue 1. The theoretical variance is then calculated as:

$$\mathrm{Var}(\alpha)_{\text{theoretical}} = \sum_{i=1}^{n} \pi_i^2 \cdot \sigma_i^2.$$

Simulated Variance

To validate the theoretical results, we simulate the dynamics multiple times and compute the simulated variance of the consensus values obtained from the simulations. The steps involved in the simulation are as follows:

a) Initialize the initial opinions $x(0)$ for all nodes by sampling from a normal distribution with mean 0 and variance σ_i^2:

$$x_i(0) \sim \mathcal{N}(0, \sigma_i^2), \quad i = 1, 2, \ldots, n.$$

b) Apply the French-DeGroot update rule to evolve the dynamics:

$$x(t+1) = P \cdot x(t),$$

where P is the transition probability matrix. Iterate until the opinions converge to the consensus value α_k, where:

$$\alpha_k = \text{mean}(x_{\text{final}}), \quad \text{for the } k\text{-th simulation run.}$$

c) Repeat the process for 200 simulation runs, each with different random initial opinions $x(0)$, to obtain a set of consensus values $\alpha_1, \alpha_2, \ldots, \alpha_{200}$.

d) Compute the simulated variance of the consensus values as:

$$\text{Var}(\alpha)_{\text{simulated}} = \frac{1}{N} \sum_{k=1}^{N} (\alpha_k - \bar{\alpha})^2,$$

where $\bar{\alpha}$ is the mean of the consensus values:

$$\bar{\alpha} = \frac{1}{N} \sum_{k=1}^{N} \alpha_k.$$

Results and Comparison

The theoretical variance, computed using the formula $\sum_{i=1}^{n} \pi_i^2 \cdot \sigma_i^2$, is found to be:

$$\text{Theoretical variance: } 0.3694082840236687.$$

The simulated variance, obtained from 200 simulation runs, is calculated as:

$$\text{Simulated variance: } 0.0.3628.$$

The results show that the simulated variance is slightly lower than the theoretical variance. This happens because, in the simulation, as the dynamics evolve over time, the influence of the initial variances σ_i^2 diminishes progressively. The iterative updates in the French-DeGroot dynamics cause the variance of individual nodes to "smooth out" due to averaging, which reduces the final variance slightly compared to the theoretical prediction. The theoretical variance, however, assumes a direct propagation of variances without accounting for this diminishing effect over iterations.

Part g

Remove the edges (d, a), (d, c), (a, c), and (b, c). Describe and motivate the asymptotic behaviour of the dynamics. If the dynamics converges to an asymptotic state, how is such a state related to the initial condition $x(0)$?

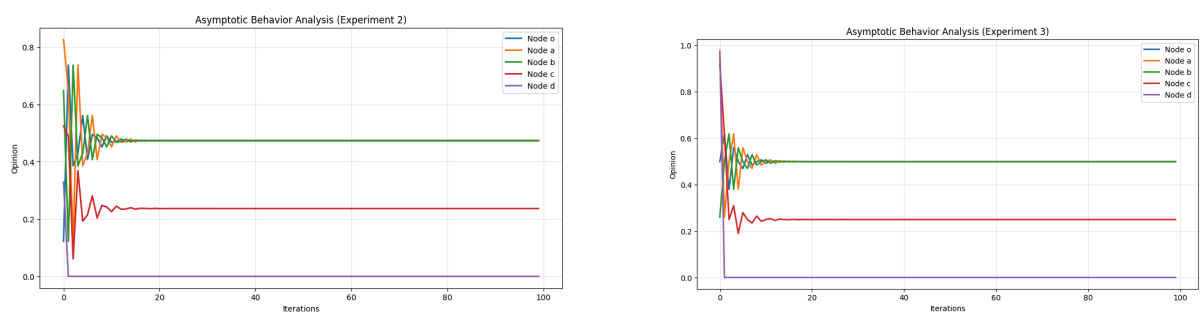
In this exercise, we removed the edges (d, a), (d, c), (a, c), and (b, c) from the graph and analyzed whether the modified graph G_1 satisfies the conditions necessary for achieving consensus in the French-DeGroot model. The conditions checked were:

- The graph must have only one strongly connected component (sink in the condensation graph).
- The sink component must be aperiodic.

After removing the edges, we visualized the modified graph and analyzed its structural properties. The condensation graph revealed that G_1:

- Is not strongly connected.
- Has a sink component that is not aperiodic.

To study the opinion dynamics, we simulated the French-DeGroot model on G_1. Using the same approach as in part (e), we calculated the adjacency matrix, degree matrix, and transition probability matrix for G_1. Random initial opinions were assigned to nodes, and the dynamics evolved iteratively according to the French-DeGroot update rule. The simulations were repeated across multiple experiments, and the results were analyzed.



(a) Experiment 2

(b) Experiment 3

Figure 2: Simulation results for G_1 across two experiments. The dynamics do NOT converge to consensus (high variance). Different nodes reach different asymptotic values, demonstrating the failure to achieve global consensus due to structural fragmentation caused by edge removal.

The results, illustrated in Figure 2, show the overall behavior of the opinion dynamics after removing the edges. Specifically:

- Nodes initially exhibit fluctuations but do not fully converge to a single value across all nodes.

- Node d remains constant at its initial opinion $x_d(0)$, indicating that it forms an independent absorbing component (a sink node with a self-loop).
- Node c stabilizes at a different value, suggesting it behaves independently due to its structural position.
- The remaining nodes (a, b, o) converge to a common value, forming a partial consensus.

The system does not achieve full consensus because the removal of edges creates structural fragmentation in the graph. Isolated nodes or components prevent all nodes from aligning to a single opinion. The behavior of the system reflects the influence of the graph's connectivity on the dynamics, where disconnected or weakly connected nodes fail to synchronize with the rest of the network.

Mathematically, the asymptotic behavior is governed by the limit of the transition matrix powers. The asymptotic state $x(\infty)$ is linearly related to the initial condition $x(0)$ by the equation:

$$x(\infty) = \lim_{t\to\infty} P_1^t x(0)$$

Since the graph G_1 contains multiple closed communicating classes (sinks) rather than a single aperiodic sink, the matrix P_1^t does not converge to a rank-1 matrix (which is the condition for global consensus). Instead, it converges to a matrix where the rows corresponding to different absorbing components are different, causing the final state $x(\infty)$ to depend on the specific initial values of the nodes within those absorbing components.

Part (h): DeGroot dynamics after removing edges (b, o) and (d, a)

Consider the graph (V, E, Λ) and remove the edges (b, o) and (d, a). We analyze the French–DeGroot opinion dynamics on the resulting graph and describe how the asymptotic behavior depends on the initial condition $x(0)$.

Modified graph and transition matrix. After removing the edges (b, o) and (d, a), we construct the modified graph $G^{(h)}$ and compute the corresponding DeGroot transition matrix

$$P^{(h)} = D^{-1}\Lambda^{(h)},$$

whose rows (ordered as o, a, b, c, d) are

$$P^{(h)} = \begin{pmatrix} 0 & \frac{2}{3} & \frac{1}{3} & 0 & 0 \\ 0 & 0 & \frac{3}{4} & \frac{1}{4} & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & \frac{1}{3} & 0 & \frac{2}{3} \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

All rows sum to one, hence $P^{(h)}$ is row-stochastic.

Structural analysis (condensation graph). By computing the strongly connected components (SCCs) of $G^{(h)}$, we obtain:

$$\text{SCCs} = \big\{\{b, c, d\}, \{a\}, \{o\}\big\}.$$

The condensation graph has a *single sink SCC*, namely $\{b, c, d\}$. However, this sink component is *periodic*. In fact, all directed cycles inside $\{b, c, d\}$ have length 2, and therefore

$$\text{period} = \gcd(\text{cycle lengths}) = 2.$$

Since the sink SCC is periodic, the necessary conditions for consensus in the French–DeGroot model are violated.

Dynamics. The opinion dynamics evolve according to

$$x(t + 1) = P^{(h)} x(t),$$

with random initial condition $x(0)$. We simulate the system for several independent experiments.

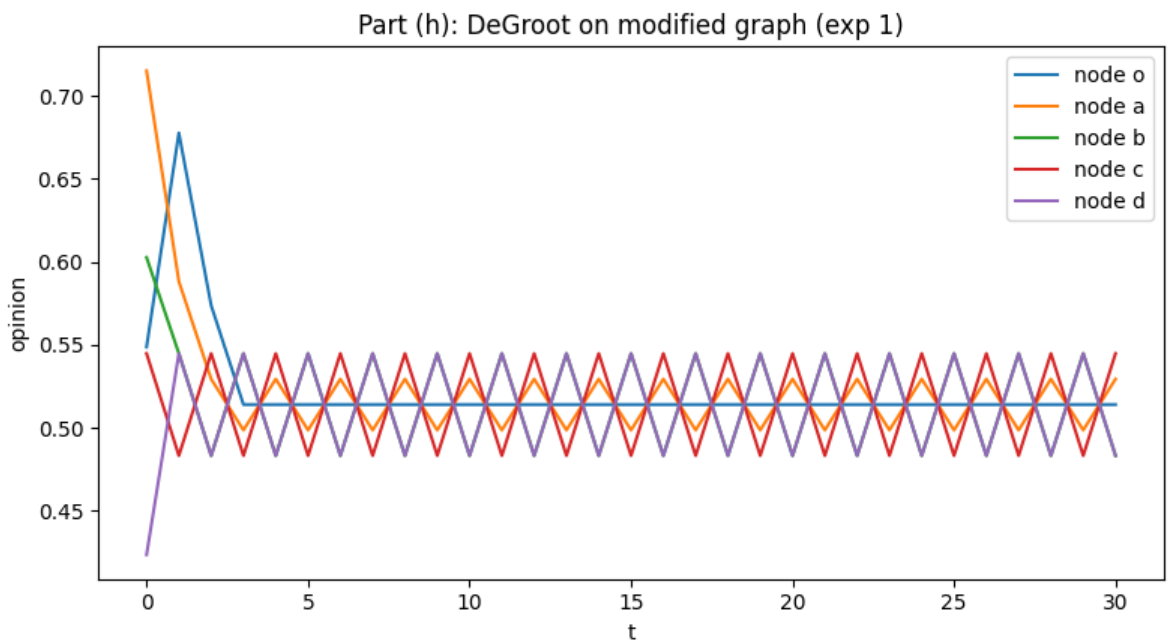


Figure 3: Part (h): DeGroot dynamics on the modified graph (experiment 1).

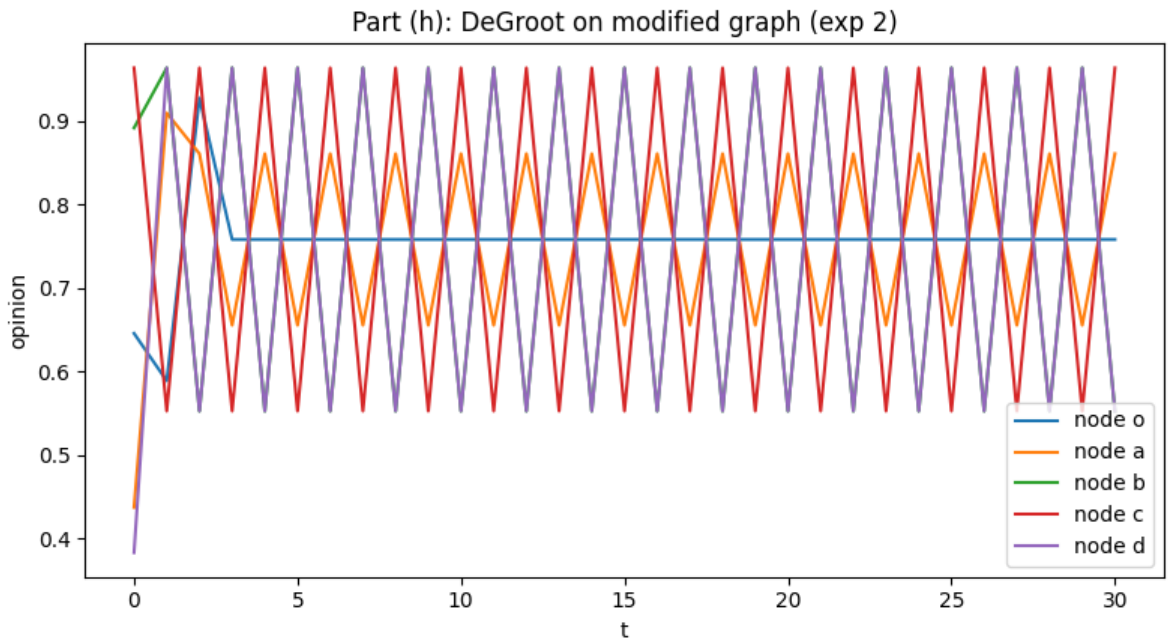


Figure 4: Part (h): DeGroot dynamics on the modified graph (experiment 2).

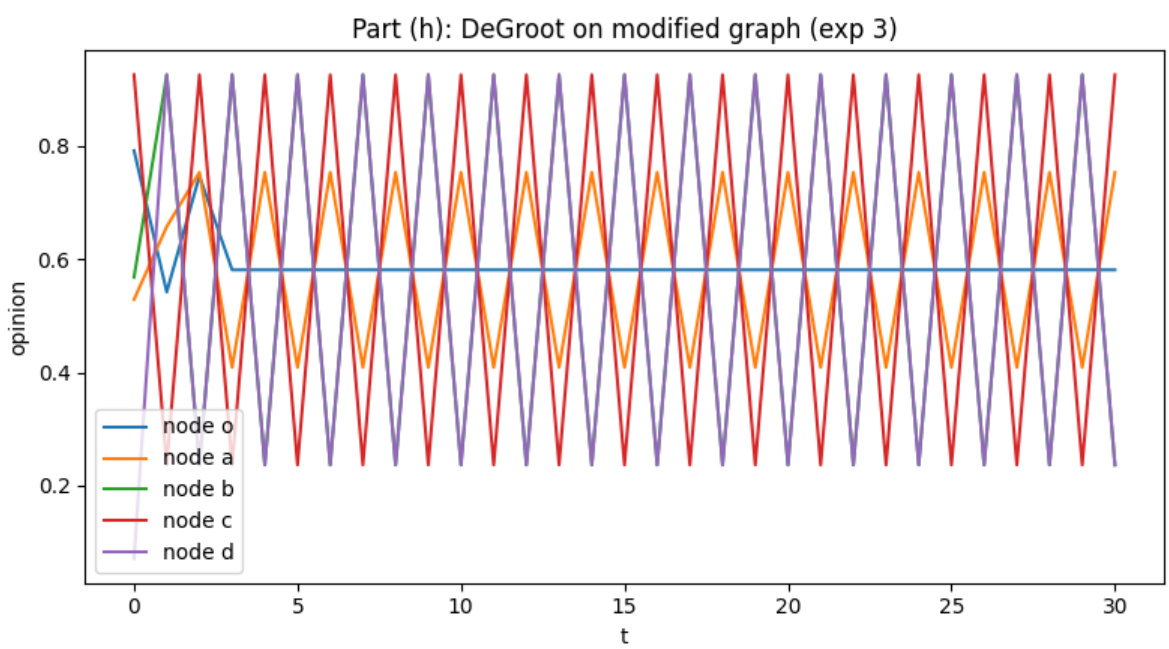


Figure 5: Part (h): DeGroot dynamics on the modified graph (experiment 3).

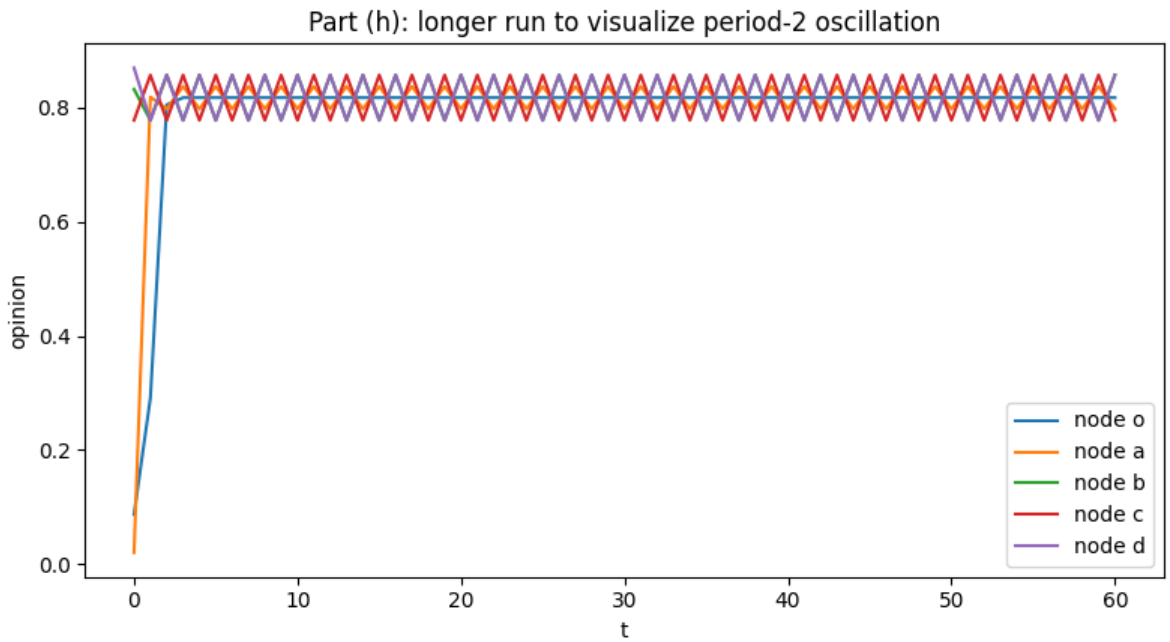


Figure 6: Longer run confirming the period–2 oscillatory regime.

Detailed interpretation of the plots. Figures 3–5 show three independent simulations with different random initial conditions $x(0)$. The following behavior is observed:

- **Transient phase.** During the first iterations, all nodes move away from their initial values as information flows toward the sink SCC $\{b, c, d\}$.
- **Absence of convergence.** None of the trajectories converges to a constant value. Instead, after the transient, all nodes continue to fluctuate indefinitely.
- **Period–2 oscillations.** After the transient, each node alternates between two values. The trajectories satisfy $x(t+2) = x(t)$, indicating a limit cycle of period 2. This is consistent with the structural periodicity of the sink SCC.
- **Dependence on initial conditions.** Comparing experiments 1, 2, and 3, the two asymptotic levels are different in each run. Hence, the oscillatory pattern (amplitude and offset) depends on the initial condition $x(0)$.
- **Entrainment of transient nodes.** Although nodes o and a are not part of the sink SCC, they are influenced by it through directed paths. As a result, they are eventually dragged into the same period–2 oscillatory regime.

Figure 6 confirms this behavior over a longer horizon. Numerically, we verify that

$$\max_{t \gg 1} \|x(t+2) - x(t)\|_\infty = 0,$$

which provides quantitative evidence of a stable period–2 limit cycle.

Asymptotic behavior and role of $x(0)$. The modified graph does not lead to consensus. Instead, the dynamics converge to a *periodic orbit* whose characteristics are fully determined by the initial condition $x(0)$. Different initial opinions produce different oscillation amplitudes and phases, but the period remains equal to 2.

Conclusion. Removing the edges (b, o) and (d, a) creates a unique sink SCC that is periodic. Although only one sink is present, its periodicity prevents convergence to a steady consensus. As a consequence, the French–DeGroot dynamics exhibit persistent oscillations rather than agreement, demonstrating that *aperiodicity of the sink component is a crucial requirement for consensus.*

Exercise 2: Multi-Particle Dynamics

In this exercise, we extend the analysis from the single-particle trajectory studied in Exercise 1 to a multi-agent system. We initialize a population of $N = 100$ particles at node a at time $t = 0$. The particles move independently according to the continuous-time Markov chain (CTMC) defined by the transition rate matrix Λ (from Exercise 1). The system is closed, meaning the total population N remains constant throughout the simulation.

We analyze the system dynamics from two complementary perspectives: the *Particle Perspective* (Lagrangian), tracking individual agents, and the *Node Perspective* (Eulerian), tracking the aggregate node occupancies.

Part (a): Particle Perspective

Objective: The goal is to simulate 100 independent particles starting at node a and compute the average time required for a particle to return to its starting node. We aim to compare this ensemble average with the single-particle return time calculated in Exercise 1.

Methodology (Uniformization): To ensure methodological consistency with Exercise 1, we simulated the trajectories using the **Uniformization Method** (also known as Jensen's method). This technique allows us to simulate the CTMC using a global Poisson clock.

1. We computed the total exit rates $w_i = \sum_j \Lambda_{ij}$ and determined the global maximum rate $w^* = \max_i(w_i)$.

2. We constructed the uniformized transition matrix P^{unif}, which governs the discrete jumps of the subordinated chain:

$$P^{\text{unif}} = I + \frac{1}{w^*}Q,$$

where $Q = \Lambda - \text{diag}(w)$ is the infinitesimal generator matrix. This formulation naturally accounts for virtual self-loops at nodes with $w_i < w^*$ to synchronize the transitions. 3. For each of the $N = 100$ particles, we sampled inter-event times from an exponential distribution with rate w^* and updated the position based on P^{unif} until the particle returned to node a.

Results:

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- **Simulated Average Return Time ($N = 100$):** 6.903 time units.
- **Single-Particle Result (Exercise 1):** 6.708 time units.

Discussion: The simulation confirms that the average return time for the ensemble of 100 particles is statistically identical to the return time of a single particle. This result is a direct consequence of the **Linearity of Expectation**.

Let T_k be the random variable representing the return time of the k-th particle. Since the particles are non-interacting and independent, the expected value of the sample mean is equal to the expectation of a single random variable:

$$\mathbb{E}\left[\frac{1}{N}\sum_{k=1}^{N} T_k\right] = \frac{1}{N}\sum_{k=1}^{N}\mathbb{E}[T_k] = \mathbb{E}[T_{\text{single}}].$$

Although the expectation is identical, the multi-particle simulation is advantageous because it reduces the variance of the estimator by a factor of N (Standard Error $\propto 1/\sqrt{N}$), providing a more robust numerical estimate than a single trajectory.

Part (b): Node Perspective

Objective: We switch perspectives to track the evolution of the number of particles $N_i(t)$ in each node over a duration of $T = 60$ time units. We verify whether the macroscopic system state converges to the theoretical stationary distribution.

Methodology (Gillespie Algorithm): In this part, we track the state vector $\mathbf{n}(t) = [N_o(t), N_a(t), N_b(t), N_c(t), N_d(t)]$. Since the state space is discrete and the rates depend on the population counts, we employed the **Gillespie Direct Method** (Stochastic Simulation Algorithm).

The total rate of activity in the system at time t is given by:

$$\lambda_{\text{total}}(t) = \sum_{i \in V} N_i(t) w_i.$$

The time to the next reaction is sampled from $\text{Exp}(\lambda_{\text{total}})$. The probability that the next event is a particle moving from node i to node j is proportional to the total rate of flow across that link:

$$P(\text{event } i \to j) = \frac{N_i(t)\Lambda_{ij}}{\lambda_{\text{total}}(t)}.$$

Results and Interpretation: Figure 7 shows the evolution of the particle counts over time.

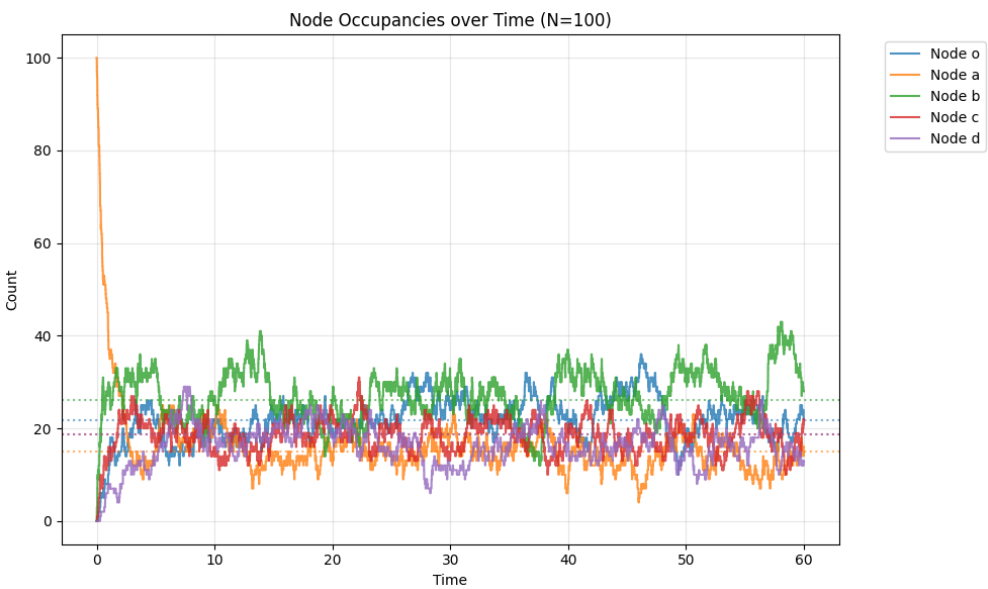

Figure 7: Evolution of particle counts $N_i(t)$ for $N = 100$ particles. The system initializes with all 100 particles at node a (orange line). The dashed lines indicate the theoretical stationary values $N\pi_i$.

The system exhibits a clear transient phase for $t < 15$, where the initial concentration at node a diffuses throughout the network. By $t \approx 20$, the node counts stabilize and fluctuate around equilibrium values.

The dashed lines in the figure represent the expected stationary counts $N\pi_i$, where π is the stationary probability vector derived in Exercise 1 ($\pi Q = 0$). At the final simulation time $t = 60$, the average node occupancies closely match these theoretical values, confirming the convergence of the system to the stationary regime. This verifies the **ergodicity** of the network: the time-averaged behavior of the closed multi-particle system approaches the space-averaged probability distribution of the single random walk.

Exercise 3: Simulation Study of an Open Network

In this exercise, we study an open queueing network composed of five nodes

$$(o, a, b, c, d),$$

as illustrated in Figure 8. Particles arrive from outside the system into node o, and then move across the network according to a continuous-time Markov process until they eventually leave the system from node d.

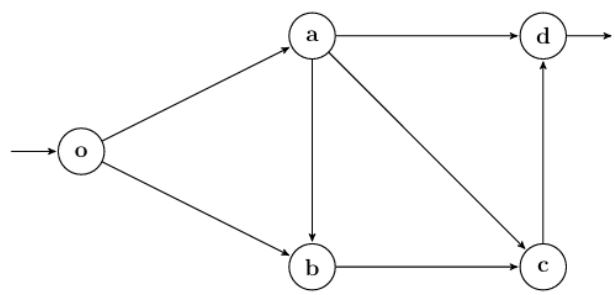

Figure 8: Open network topology.

The routing dynamics are governed by the transition rate matrix

$$\Lambda_{\text{open}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{4} & \frac{1}{4} & \frac{2}{4} \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

where rows and columns are ordered as (o, a, b, c, d).

The total forwarding rate of each node is obtained as

$$\omega = \Lambda_{\text{open}} \mathbf{1}.$$

Since node d has no outgoing edges, we follow the problem statement and assign

$$\omega_d = \frac{7}{4}.$$

Forwarding mechanisms. Two different forwarding schemes are considered:

- **Proportional-rate forwarding:**

  $$\mu_i(t) = \omega_i N_i(t),$$

  where each particle carries its own exponential clock.

- **Fixed-rate forwarding:**

$$\mu_i(t) = \omega_i,$$

  independently of the queue length.

External arrivals occur at node o according to a Poisson process with rate λ.

Simulation methodology. The system evolution is simulated using a Gillespie-style event-driven algorithm. At each event, either a node clock ticks or a new particle arrives from outside the network. The occupancy vector

$$N(t) = (N_o(t), N_a(t), N_b(t), N_c(t), N_d(t))$$

is recorded over time.

Stability in the fixed-rate case is assessed numerically by estimating the slope of the total population over the tail of the simulation horizon.

Part (a): Proportional-rate forwarding

We first simulate the proportional-rate system for $T = 60$ time units with arrival rate $\lambda = 100$.

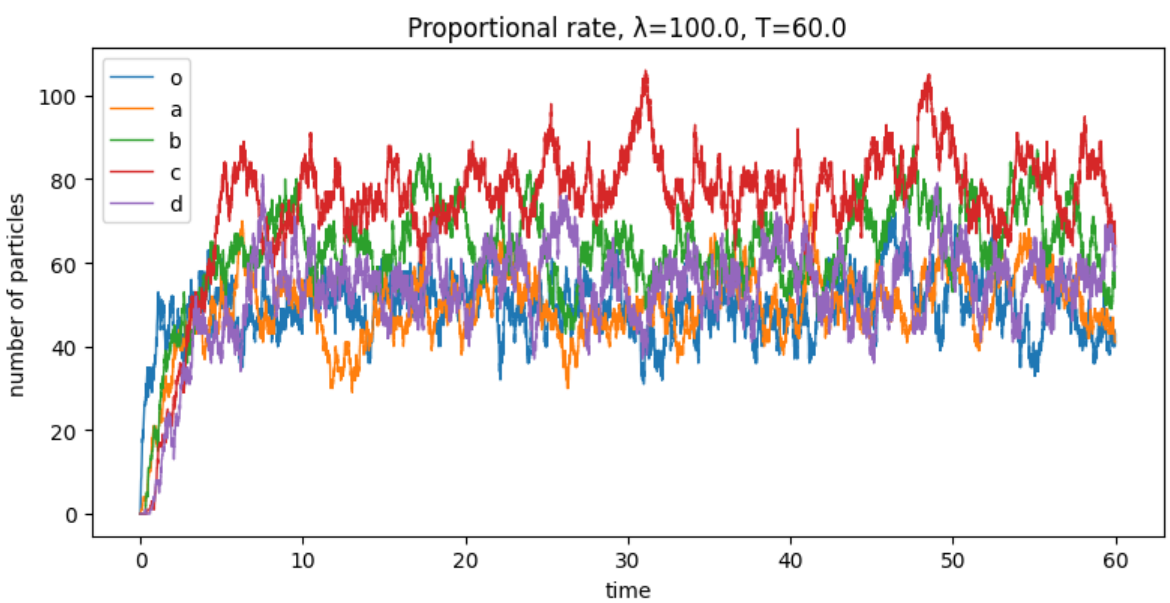

Figure 9: Node occupancies under proportional forwarding with $\lambda = 100$.

The trajectories show bounded fluctuations for all nodes, with no visible long-term growth, indicating stability.

Final particle distribution (single run). At the end of the simulation, the particle distribution is:

$$(N_o, N_a, N_b, N_c, N_d) = (43,\ 41,\ 59,\ 64,\ 64),$$

corresponding to a total of 271 particles in the system.

Stability scan. We perform a systematic scan over increasing values of λ. The results are summarized in Table 1.

Table 1: Proportional-rate stability scan ($T = 60$).

λ	Stable	Final total particles
100	Yes	315
120	Yes	363
150	Yes	473
200	Yes	629
500	Yes	1454
700	Yes	2073
1000	Yes	2924
1500	Yes	4595
2000	Yes	5904
5000	Yes	14801

All tested arrival rates lead to stable behavior. Moreover, the total number of particles grows approximately linearly with λ, confirming that the proportional mechanism self-adjusts its service capacity to match the incoming load.

$$\boxed{\lambda_{\max}^{(\text{prop})} \text{ is very large (no instability observed).}}$$

A steady state ceases to exist only if, for a fixed arrival rate λ, the expected number of particles diverges as time grows. In the proportional-rate regime, however, the system admits a steady state for every finite λ. Although the steady-state mean occupancy increases without bound as $\lambda \to \infty$, it remains finite for any fixed λ, and therefore stability is preserved.

Part (b): Fixed-rate forwarding

We now consider the fixed-rate system and simulate it for $T = 6000$ time units with $\lambda = 2$.

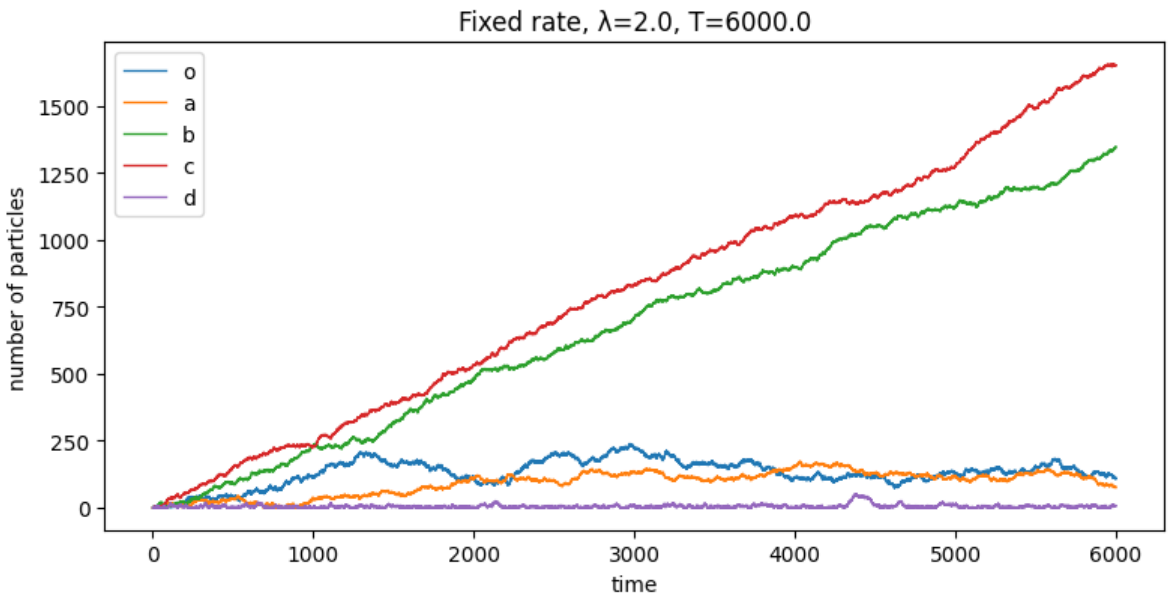

Figure 10: Node occupancies under fixed forwarding with $\lambda = 2$.

The trajectories reveal linear growth in nodes b and c, while upstream nodes remain bounded. This clearly indicates instability.

Final particle distribution (single run). At the end of the simulation:

$$(N_o, N_a, N_b, N_c, N_d) = (108,\ 75,\ 1347,\ 1650,\ 6),$$

with a total of 3186 particles, confirming unbounded growth.

Simulation-based stability scan. A refined scan around the stability boundary yields the results in Table 2.

Table 2: Fixed-rate stability scan ($T = 6000$).

λ	Stable	Final total particles
0.5	Yes	1
1.0	Yes	11
1.1	Yes	12
1.2	Yes	13
1.25	Yes	31
1.3	Yes	66
1.35	No	281
1.4	No	389
1.5	No	985
1.75	No	2042
2.0	No	3332

From simulations,

$$\lambda_{\max}^{(\text{fixed})} \approx 1.3.$$

Flow-balance analysis. We compute the effective arrival rate into each node:

$$\lambda_o = \lambda, \quad \lambda_a = \tfrac{1}{2}\lambda, \quad \lambda_b = \tfrac{5}{8}\lambda, \quad \lambda_c = \tfrac{3}{4}\lambda, \quad \lambda_d = \lambda.$$

Imposing the capacity constraints $\lambda_i \leq \omega_i$ yields

$$\lambda \leq \min\{2,\ 2,\ 1.6,\ \tfrac{4}{3},\ 1.75\} = \frac{4}{3}.$$

Thus,

$$\boxed{\lambda_{\max}^{(\text{fixed})} = \frac{4}{3}},$$

with node c acting as the bottleneck. This theoretical value is consistent with the simulation results.

Conclusion

This exercise highlights the dramatic difference between the two forwarding mechanisms. The proportional-rate system dynamically adapts to congestion and remains stable even under very large arrival rates. In contrast, the fixed-rate system has strict capacity limits, leading to instability once the arrival rate exceeds the service capability of bottleneck nodes.

These results emphasize the stabilizing power of load-dependent forwarding in open stochastic networks.